

CENTRALIZING TASK-BASED APPROACH TO QUANTUM NETWORK CONTROL

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Background

- The main goal of a quantum network is to distribute entanglement. This is fundamentally different from classical networks.
- Many proposed network stacks have been proposed. Mainly, focusing on hierarchical layered stacks [1,2].
- Hierarchical architectures suffer from common limitations:
 - Layered processing increased latency, degrading entanglement fidelity.
 - Entanglement spans layers, thus tying the resource to one layer limits usability.
 - Transient states do not warrant full hierarchical tracking.
 - Decoherence time budgets make task-processing optimization obligatory.
- A. Pirker et. al., proposed a resource-centric task-based approach to quantum network control (Saga framework) [3].
- The saga framework is agnostic of control plane implementation (distributed or centralized), and is not affected by the shortcomings of layered architectures.

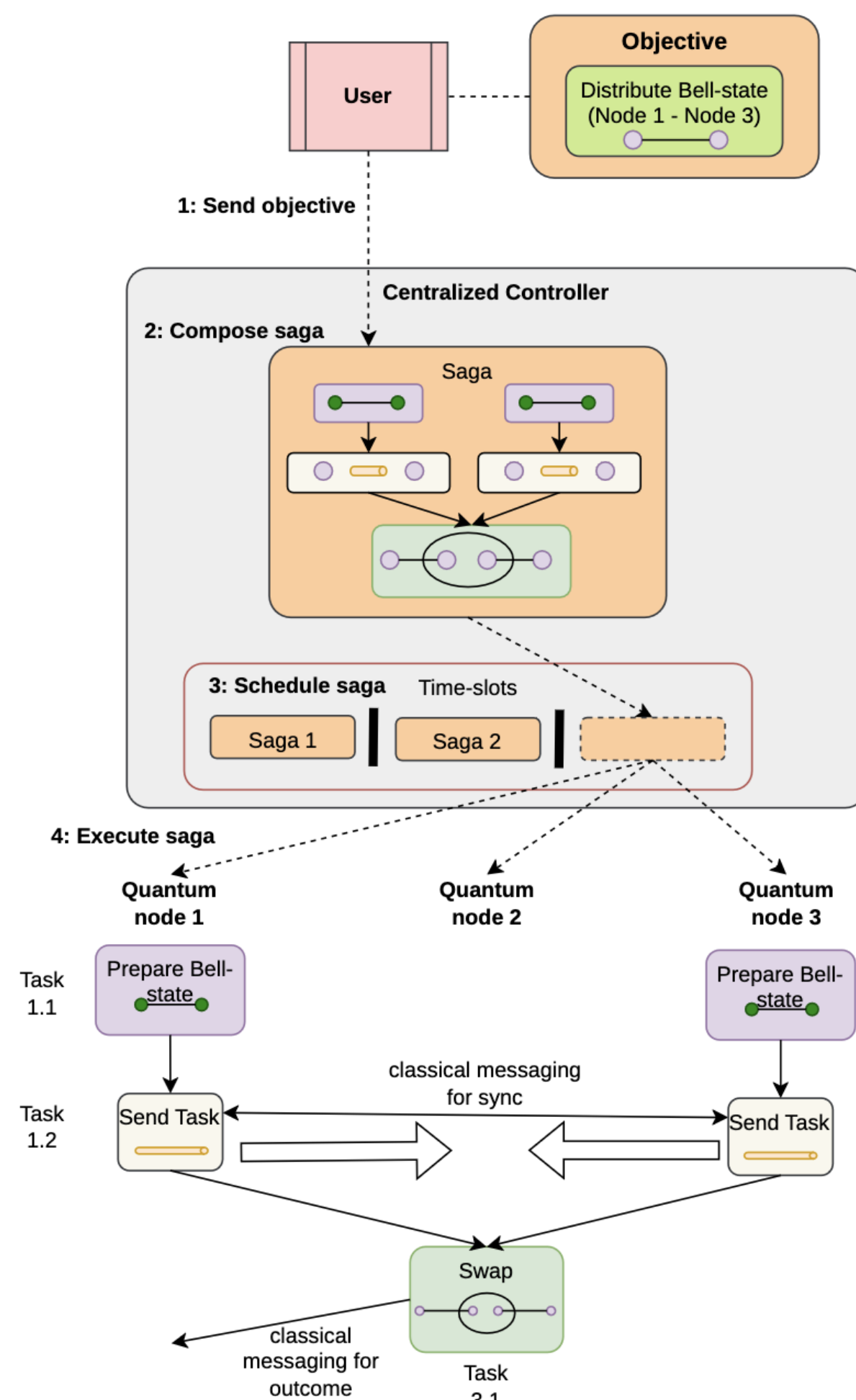


Fig. 1: Saga Architecture

Architecture

- The architecture derives distributed workflows (*sagas*) to accomplish *objectives* through *tasks*.
- Control flow for the saga architecture.
 - The user sends the objective to the central controller.
 - The centralized controller checks the currently available resources in the network and derives a saga by determine the path and resource availability of each selected node.
 - The central controller schedules the saga on the network by delaying until a valid timeslot is found.
 - The saga is transmitted to each involved node to begin executing tasks.

Simulation Methodology

- Implemented the saga architecture in SeQUeNCe where each node has memories equal to its graph degree.
- We sample {100,1000, 10000} objectives from a traffic matrix.
- Each objective is given a priority sampled from {0,1,2}.
- The arrival rate of the objectives is exponentially distributed.
- Path calculation uses Dijkstra's shortest path algorithm
- CNOT error probability is 0.99 and measurement error probability is 0.995.
- Delay of the objective is scaled by the priority and timeslot size.
- Evaluated on grid, caveman, bottleneck, and star topologies.

Results

- Fig. 3 shows the normalized proportion of objectives in a node which caused a scheduling conflict. We notice that the hub node in star topology is uncongested. This is due to the node having memories equal to its degree.
- Fig. 4 suggest that for increasing queue sizes, the transition from low to high delays shifts linearly.
- Fig. 5 shows a strong correlation with the number of hops and fidelity, while showing no correlation to priority.
- Fig. 6 suggests that the centralized saga architecture performs well under high congestion for various network topologies.

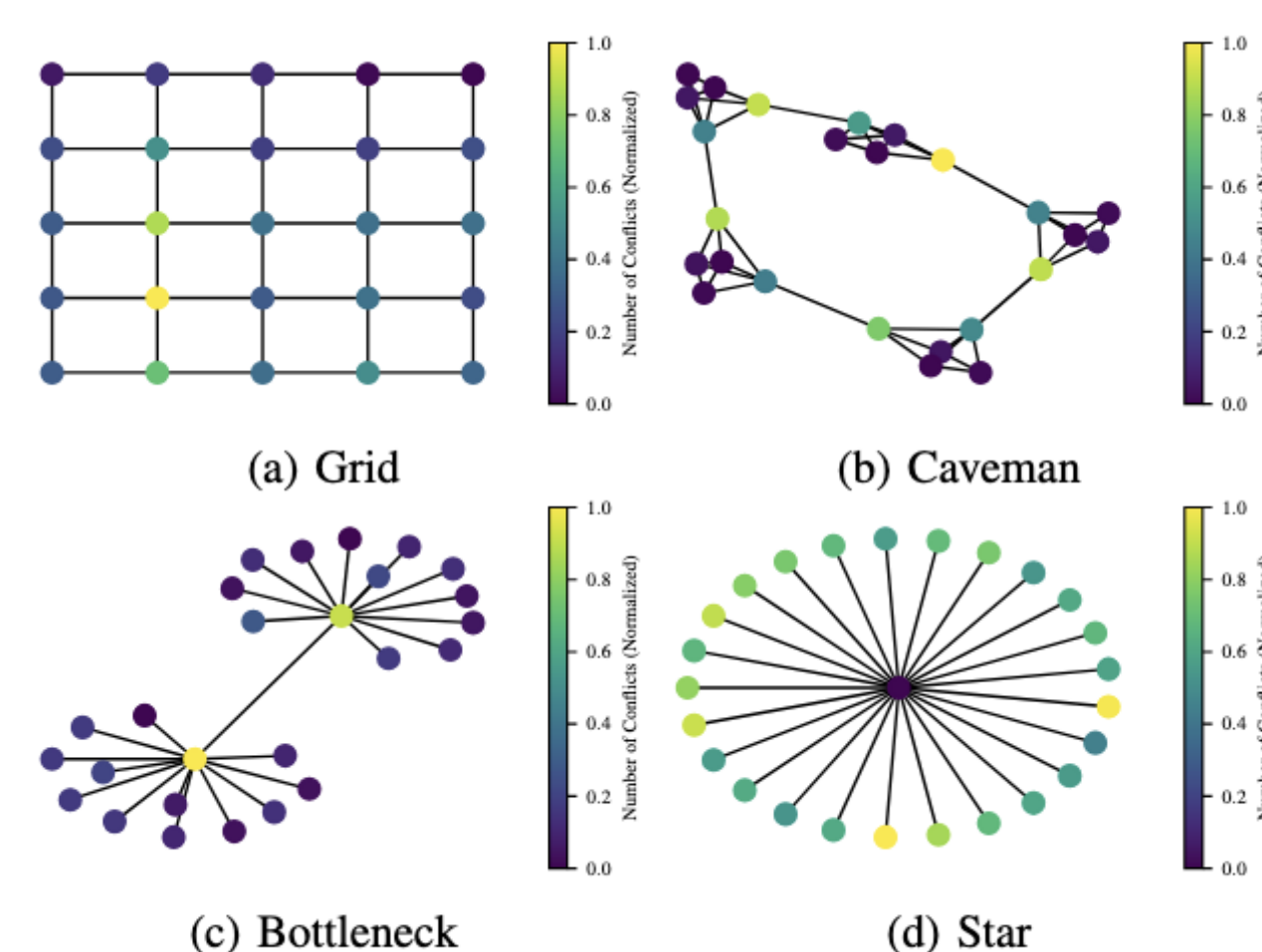


Fig. 3: Congestion for each node of the topologies for 10000 objectives

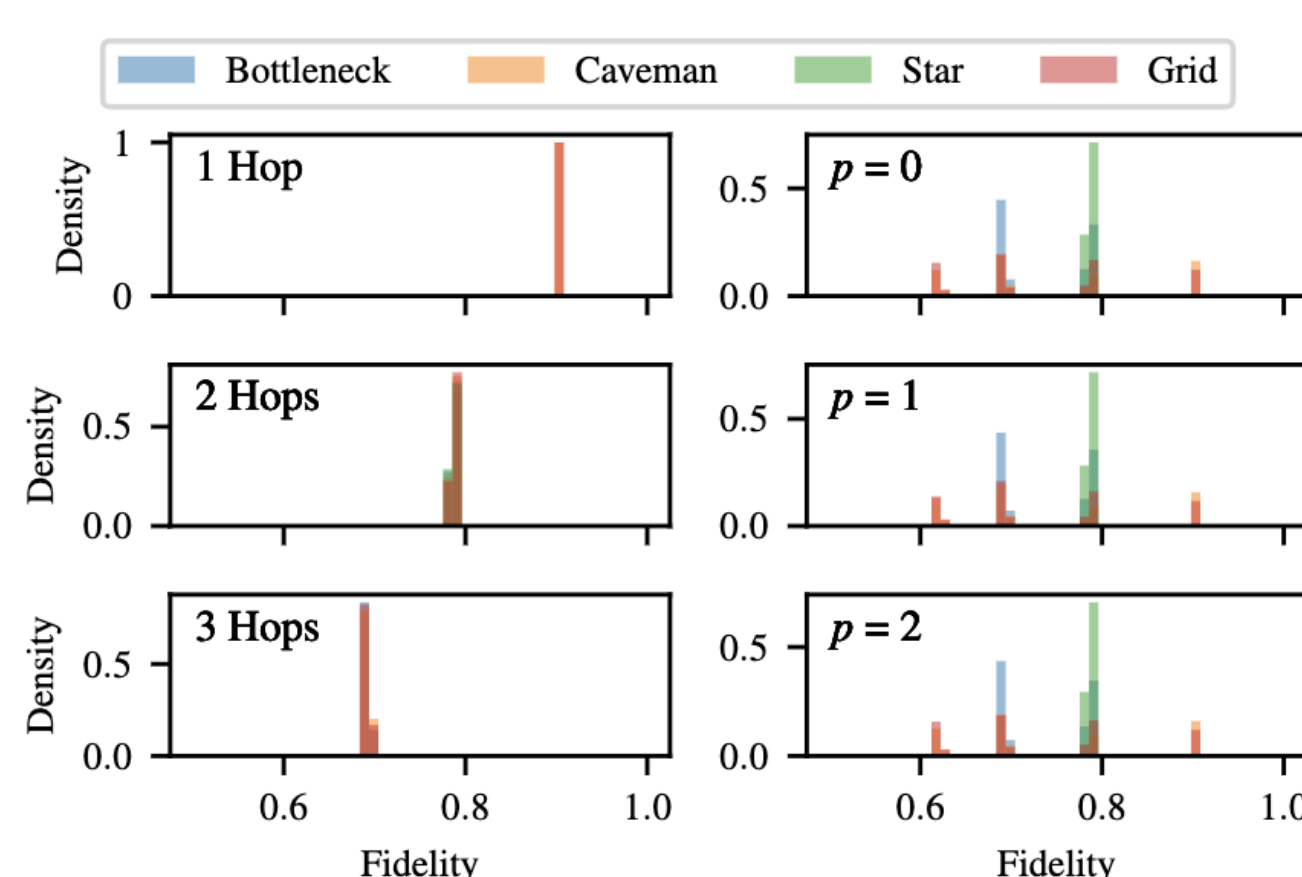


Fig. 5: Entanglement fidelity distributions separated by priority (right) and path length (left)

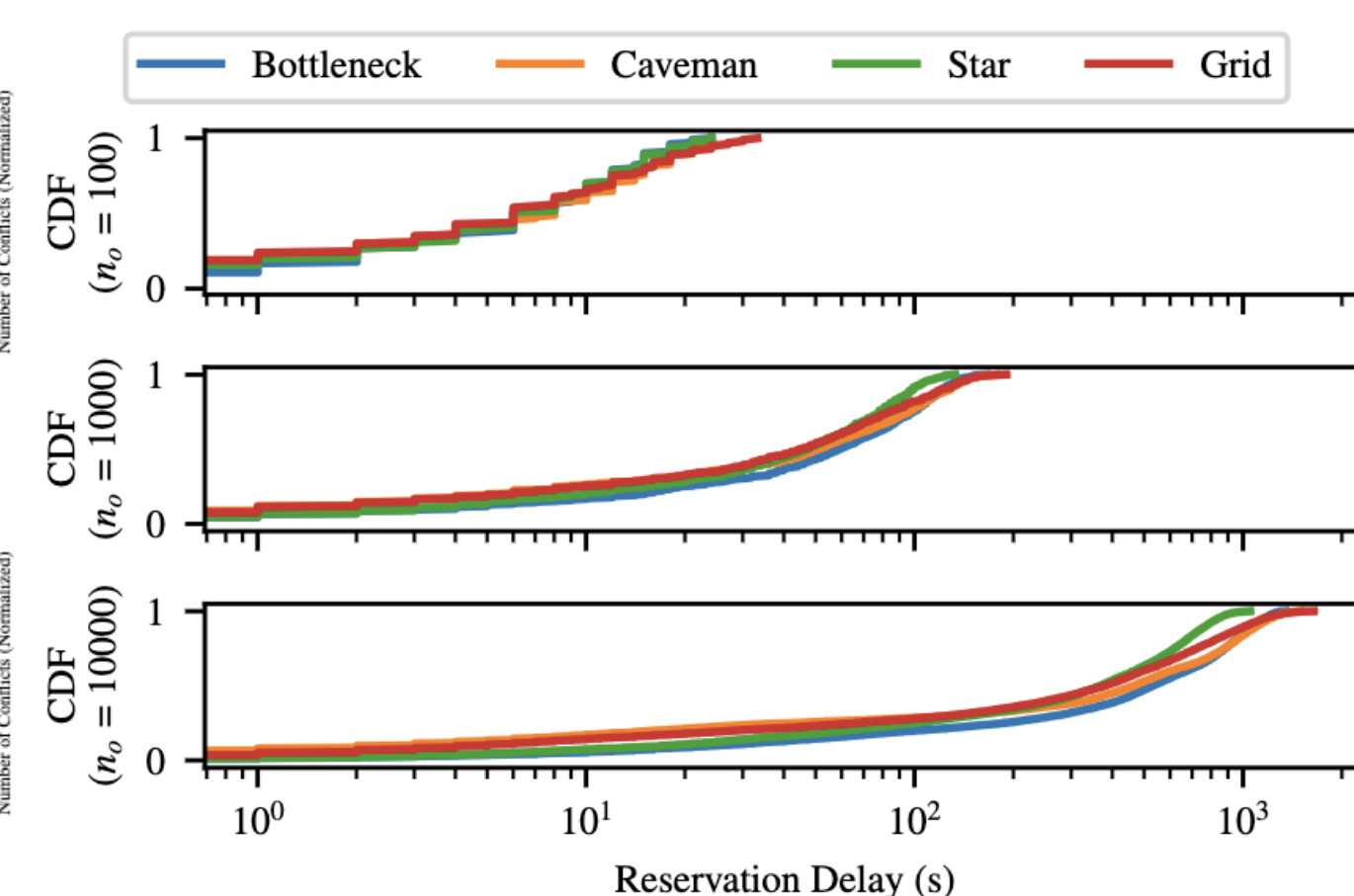


Fig. 4: CDF of objective delay for each request size separated by topology.

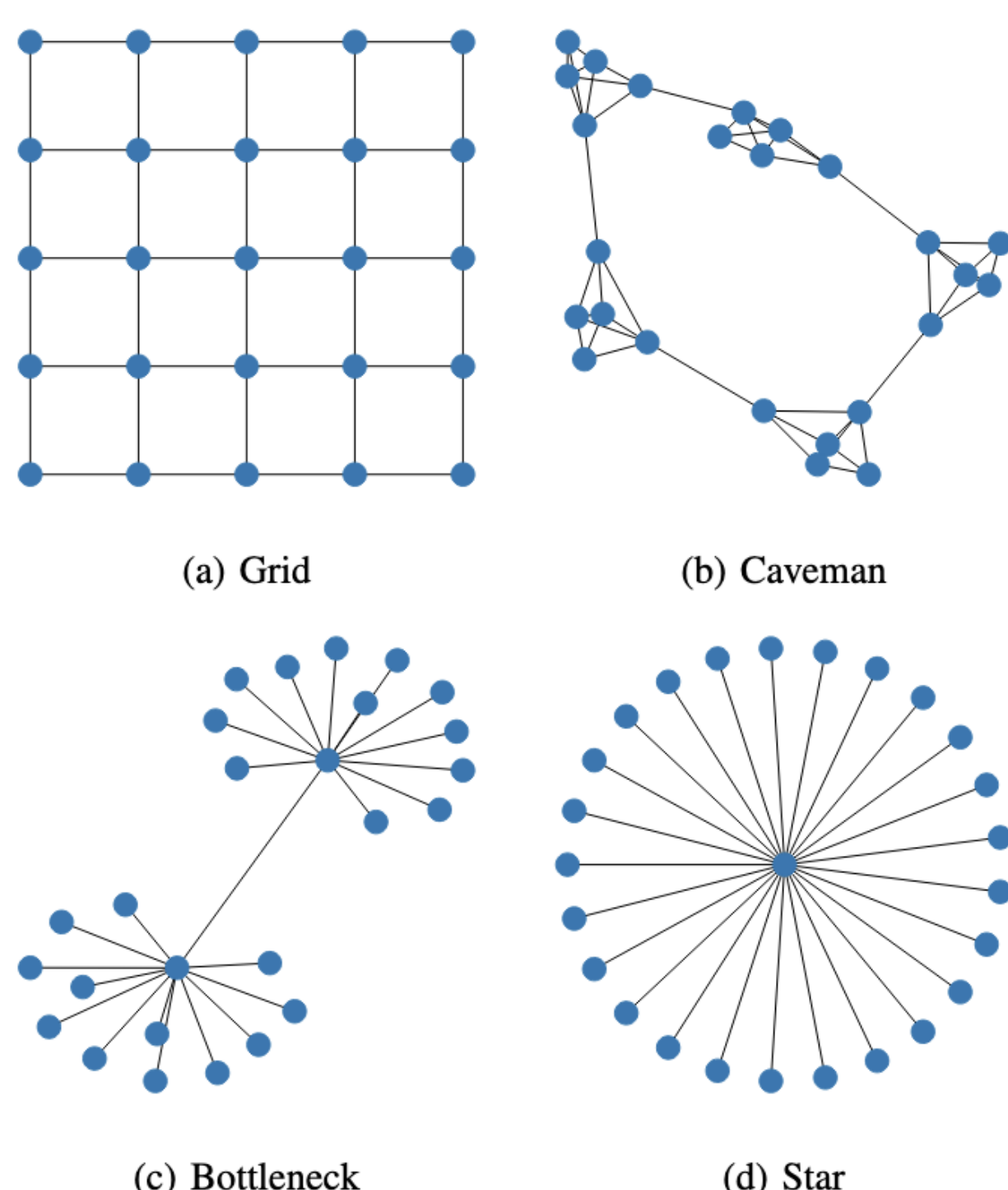


Fig. 2: Topologies

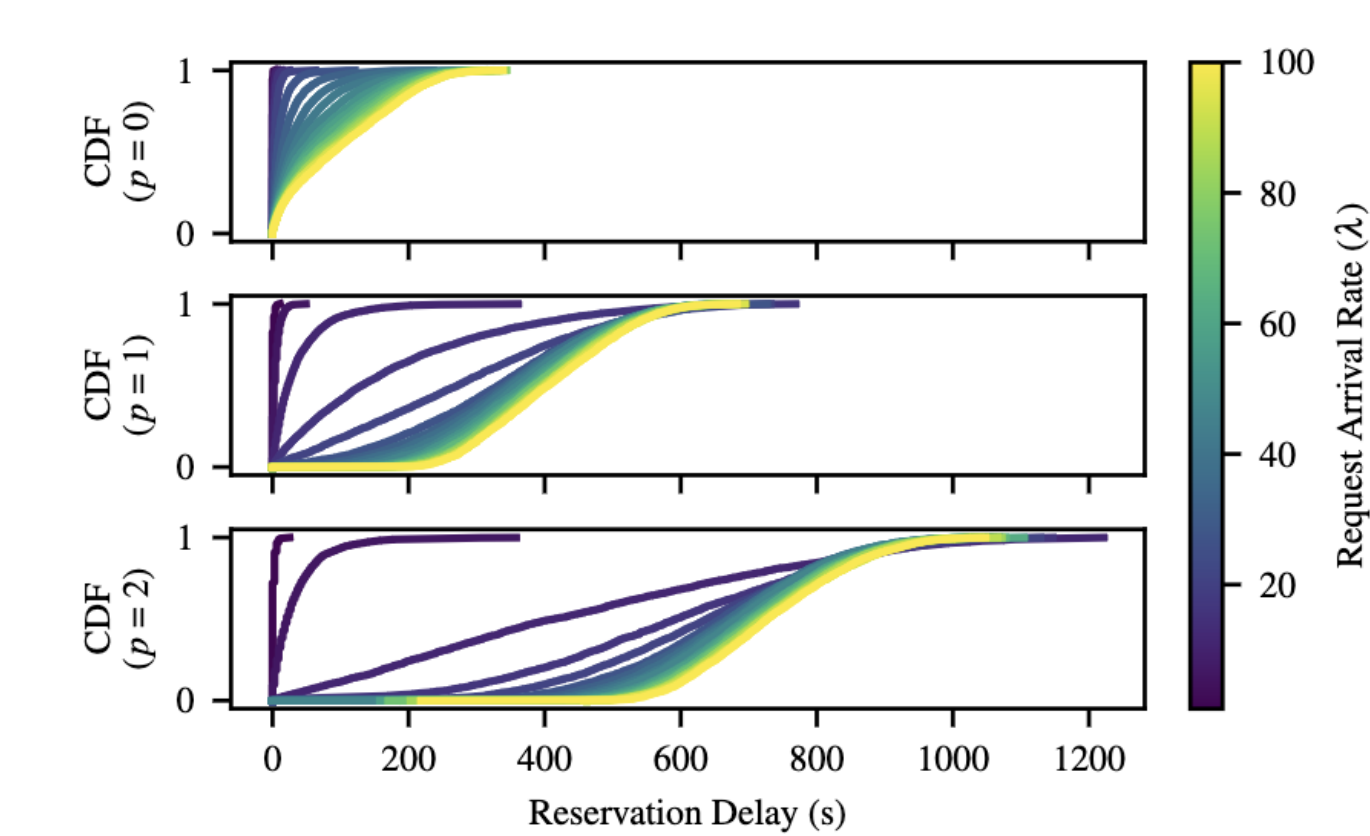


Fig. 6: CDF of objective delay with increasing arrival rate.

[1] R. Van Meter, T. D. Ladd, W. J. Munro, and K. Nemoto, "System design for a long-line quantum repeater,"

[2] A. Dahlberg, M. Skrzypczyk, T. Coopmans, L. Wubben, F. Rozpundineddek, M. Pompili, A. Stolk, P. Pawełczak, R. Knegjens, J. de Oliveira Filho, R. Hanson, and S. Wehner, "A link layer protocol for quantum networks,"

[3] A. Pirker, B. Munoz, and W. Dür, "A resource-centric, task-based approach to quantum network control,"